

multiplication

length of numbers
 $n = \max \# \text{ of digits of } x, y$

gradeschool: $\Theta(n^2)$

$x \cdot y$

$x = \boxed{1234} \boxed{5678}$

$\overset{sm}{\text{split multiply}}(x, y, n) :$

if $n=1 :$

return $x \cdot y$

else :

$m = n/2$

$a = x/10^m ; b = x \bmod 10^m$

$c = y/10^m ; d = y \bmod 10^m$

$\left[\begin{array}{l} e = sm(a, c, m) \\ f = sm(b, d, m) \\ g = sm(b, c, m) \\ h = sm(a, d, m) \\ \text{return } 10^{2m}e + 10^m(g+h) + f \end{array} \right]$

we can multiply
by 10^n in $O(n)$ time
(just shift left
and add n 0s)

with table:

- (1)
- (2) recurrence
- (3) closed form

$$\begin{aligned} x \cdot y &= (10^m a + b)(10^m c + d) \\ &= 10^{2m} \underline{ac} + 10^m \underline{ad} + 10^m \underline{bc} + \underline{bd} \end{aligned}$$

$$T(1) = 1$$

$$T(n) = 4T\left(\frac{n}{2}\right) + \Theta(n)$$

$$= \Theta(n^2)$$

$$T(n) = 3T\left(\frac{n}{2}\right) + \Theta(n)$$

$$\begin{aligned}
 &\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\
 &\boxed{bc + ad} = ac + bd - \boxed{(a-b)(c-d)} \\
 &= \cancel{ac} + \cancel{bd} - \cancel{ac} + \cancel{ad} + \cancel{bc} - \cancel{bd}
 \end{aligned}$$

Divide + Conquer

array A of length n indexed starting at 1
 $\text{Quicksort}(A[1..n])$:

if $n > 1$:

choose pivot element $A[p]$ ^{random}

$r = \text{Partition}(A, p) \in O(n)$

w/ table

$\left[\begin{array}{l} \text{quicksort}(A[1..r-1]) \\ \text{quicksort}(A[r+1..n]) \end{array} \right]$

	1	2	3	4	5	6	7	8	9	10	11	12	13
Input:	S	O	R	T	I	N	G	E	X	A	M	P	L
Choose a pivot:	S	O	R	T	I	N	G	E	X	A	M	P	L
Partition:	A	G	O	E	I	N	L	M	P	T	X	S	P

$\text{Partition}(A, p)$:

puts everything smaller than p
on L and bigger than p on R
return index of p

assume Partition runs in $O(n)$

w/ table: what is runtime?

recurrence: ^{worst-case}

$$\begin{aligned} \rightarrow T(1) &= 1 \\ T(n) &= n + T(0) + T(n-1) \end{aligned}$$



